

The functional and structural impacts of the G350D mutation on dynamin-related protein 1

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Abstract:

Dynamin-related protein 1 (Drp1) is a key regulator of mitochondrial fission. It assembles into helices that encircle and constrict the mitochondria upon GTP hydrolysis, eventually leading to fission. Mutations in Drp1 can disrupt fission and cause neurodevelopmental and neurodegenerative diseases, though the underlying mechanisms are not fully understood. This study introduces the G350D mutation into Drp1 using bacterial expression constructs to explore its structural and functional impact on the protein. The G350D mutation is located in the Middle Domain region of the protein at one end of the L1N loop - a known hotspot for mutations disrupting assembly. However, the specific influence of this loop on assembly is unknown. Our results revealed that the G350D forms more hexamer structures than the WT in solution, indicating stabilization of larger fission inhibitory complexes. Furthermore, the mutation exhibits increased rates of enzymatic activity (4.15 min⁻¹) compared to the WT (0.53 min⁻¹). Negative stain electron microscopy demonstrated that the WT was able to assemble into spirals in the presence of GMP-PCP, whereas the G350D mutation was unable to do so, indicating its assembly defectiveness. Analyzing the effects of specific mutations like G350D is crucial for comprehending Drp1's function and the roles of specific sites like the L1N loop, providing insights into how neurodevelopmental and neurodegenerative diseases occur and offering potential treatment options. Continuing to research mitochondrial dynamics is vital to fully understand the role that mitochondria play in the body and improve human health.

Introduction and Background:

Mitochondria are responsible for producing ATP (adenosine triphosphate) through cellular respiration. Some cells in the body have higher energy demands for mitochondrial function; therefore, it is essential to generate more ATP in these cells. Mitochondrial dynamics, dictated by the opposing processes of fusion and fission, change the morphology of the mitochondria impacting ATP production and distribution. (1) Fusion is when two mitochondria encounter each other, merge, and become one. Mitochondrial fission is the process in which a singular mitochondria can split and develop into two or more mitochondria. In order for these cells to maintain healthy function, a balance between mitochondrial fission and fusion is crucial. (2)

Fission is regulated by dynamin-related protein 1 (Drp1), a large GTPase that assembles into helices that encircle and constrict the mitochondrial membrane upon GTP hydrolysis, eventually leading to mitochondrial fission. (3-4) This protein is composed of four distinct domains that allow Drp1 to mediate this division event. The GTPase domain hydrolyzes GTP as the core enzymatic region. The middle domain (MD) and GTPase effector domain (GED) form an extended stalk that controls the self-assembly of Drp1 into a multimeric helical structure. Lastly, the variable domain (VD) is responsible for lipid sensing on the surface of the mitochondria and regulates the stalk regions to influence when and where Drp1 assembles in the cell.

Self-assembly is essential for allowing for mitochondrial fission to occur. Without proper assembly, Drp1 cannot form the helical structures necessary to constrict the mitochondrial membrane through GTP hydrolysis, resulting in impaired fission. Furthermore, the lack of assembly prevents the enhanced activity of Drp1 which is thought to drive membrane division.

Mutations in the MD of Drp1 have been found to disrupt protein assembly, as this region plays a vital role in controlling the self-assembly of Drp1. Within this region, a critical loop structure has been identified as a hotspot for mutations—both patient-derived and synthetic—that interfere with assembly. This loop (termed L1N) is crucial for stabilizing interactions with other parts of the protein and stimulating assembly activities. However, the specific mechanism by which the L1N loop region influences assembly and is disrupted by mutations is still not fully understood.

There are numerous synthetic mutations that have been shown to influence assembly. For instance, a study using yeast as a model organism discovered that the Dnm1G385D mutation impairs Dnm1's ability to self-assemble into higher-order structures, which are essential for mitochondrial fission in yeast cells. (5) Similarly, the synthetic G363D mutation in Drp1, studied using CHO cells, disrupts the higher-order assembly of Drp1 complexes at mitochondrial surfaces, and assembly-stimulated GTPase activity is impaired *in vitro*. (6) Patient mutations in this loop are also associated with poor clinical outcomes. For example, the patient mutation p.G350R in Drp1 altered the morphology of the mitochondria and caused death in transgenic *Drosophila* models, demonstrating a significant effect on mitochondrial function. (7) Another study identified a de novo missense mutation, p.(Gly362Asp), in Drp1 from patient-derived cells, which is linked to refractory epilepsy. Similar to synthetic mutations in the L1N loop, this mutation disrupts protein oligomerization, thereby also affecting protein function. (8)

Patient mutations in the Drp1 protein and other proteins that regulate mitochondrial fission and fusion can cause neurodegenerative diseases. (9) Although it is known that mutations in the Drp1 protein cause neurodevelopmental and neurodegenerative diseases, the method by which the mutations progress to the diseases is still being researched. Furthermore, even though synthetic mutations in the Drp1 protein have helped determine the functional role of domains and

patient mutations have been studied, the exact way in which Drp1 activity and mitochondrial fission are impaired is uncertain.

In this study, we are introducing the G350D mutation in Drp1. This mutation has been incorporated into bacterial expression constructs, allowing us to express the mutated Drp1 protein itself. By doing so, we aim to investigate how the G350D mutation influences the biochemical and structural function of Drp1. We will compare the mutated Drp1 with a wild-type control sample to characterize the functional differences caused by the mutation. This research is crucial as it can increase our understanding of how Drp1 mutations can lead to the development of neurodegenerative diseases. Moreover, these studies will provide further insights into the structure and function of the L1N loop in the protein.

We hypothesize that the G350D mutation will cause an assembly defect since the mutation is located in the MD of the protein. Additionally, this mutation is located on one end of the L1N loop (Figure 1) which has been the hotspot of mutations associated with assembly defects.

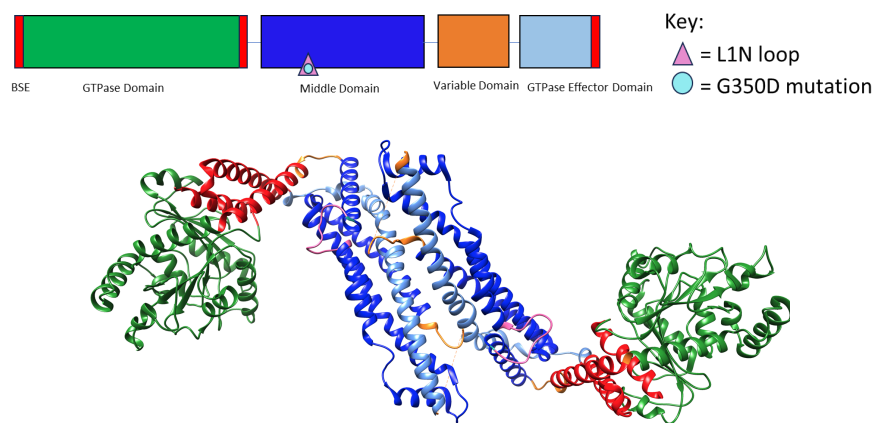


Figure 1. Diagram depicting the domains of Drp1 chains A and B. The green region represents the GTPase domain, the red region represents the BSE, the dark blue represents the MD, and the light blue represents the GED. The L1N loop is indicated by a pink triangle, with the G350D mutation, located in this loop, marked by a teal circle.

Methods:

Our approach for this study was to focus on protein expression in BL21 cells after validating the sequence change in the DNA through sequencing.

Location:

Most assays and methods listed below were completed in the Mears lab. The negative stain EM was conducted in an EM facility at the Cleveland Center for Membrane and Structural Biology (CCMSB, <https://case.edu/medicine/ccmsb/>).

Bacterial Culture:

Our first step was to express our plasmid DNA (deoxyribonucleic acid) with the G350D mutation in DH5-Alpha competent *E. coli* cells. We used the High-Efficiency Transformation Protocol (C2987H/C2987I). We used these specific cells due to their ability to generate large quantities of cloned DNA. We followed the established protocol but omitted steps 6, 7, and 9, which included the use of the SOC mixture. After colonies had grown in the selection plate, one colony was chosen and placed into a culture tube containing Luria broth (LB) and ampicillin. This culture tube was then placed in an incubated shaker overnight at 37°C to start the bacterial culture.

Plasmid DNA purification:

We completed the DNA purification to isolate the plasmid DNA in the bacterial culture from genomic DNA, proteins, ribosomes, and bacterial cell wall components using the ZymoPURE Plasmid Miniprep Kit from Zymo Research.

Sequencing:

We sent the purified plasmid DNA sample to Azenta Life Sciences for Sanger Sequencing to validate if the sequence change for our mutation was made. To prepare the sample, we added 10 µL of DNA from our sample into each sequencing tube. Then, we added

1.6 μL of the 740F primer into two of the sequencing tubes and 1.6 μL of the Y322A primer into the other two sequencing tubes. Finally, we added 3.3 μL of H_2O to each tube. The sequence change for our mutation was confirmed.

BL21 Bacterial Culture:

After confirming our sequence, we conducted bacterial transformation using BL21 (DE3) competent *E. coli* cells. These cells were chosen because our goal is to express protein. These cells have a T7 RNA polymerase gene which facilitates protein expression. Furthermore, they respond well to IPTG (isopropyl β -D-1-thiogalactopyranoside) induction, which triggers the T7 RNA polymerase to begin expressing the target protein. We followed protocol C2527 for this transformation. We omitted steps 6, 7, and 9, which involved using the SOC mixture. After allowing the colonies to grow, we selected one and transferred it into a culture tube containing Luria broth (LB) and ampicillin. The tube was then placed in a shaking incubator overnight at 37°C to start the bacterial culture.

IPTG Induction:

To start protein expression in the BL21 cells, we performed IPTG Induction. We prepared two 1000 mL (milliliters) flasks of media and added 1 mL of ampicillin to both flasks. Next, we added 20 mL of our BL21 bacterial culture to each flask. The flasks were then shaken for 3 hours at 37°C. After shaking, we allowed the flasks to cool before adding 1 mL of IPTG to each flask. They were then placed in the incubated shaker at 18°C for 24 hours.

Pelleting Cells:

The flasks were removed from the incubated shaker and transferred into two 750 mL Polypropylene Centrifuge Bottles from Beckman. Both bottles were balanced on a scale to ensure a difference of 0.01 grams or less. They were then centrifuged at 4300 RPM (Revolutions

Per Minute) for 20 minutes. After centrifugation, the supernatant was removed, leaving a pellet at the bottom of the bottles. The remaining liquid from the flasks was transferred into the Centrifuge Bottles and centrifuged again at 4300 RPM for 10 minutes. Again, the supernatant was discarded. The cell pellet remaining at the bottom of the Centrifuge Bottles was resuspended with 25 mL of 1xPBS. The resuspended solution was then transferred into two 25 mL conical tubes. These tubes were centrifuged for fifteen minutes at 4750 RPM. After centrifugation, the supernatant was discarded. The conical tubes containing the pellet were stored at -80°C until protein purification.

Protein Purification:

To purify our mutated protein sequence, we followed the established protocol. ([10](#))

GTPase Assay:

To measure the GTPase activity of the mutated sequence, we followed the established protocol.

Negative Stain Electron Microscopy:

To visualize the oligomerization capabilities of the mutated sequence, we performed the established protocol regarding the Qualitative Assessment of Drp1 Oligomerization by EM. ([10](#))

Mass Photometry:

The standard protocol was used to measure light scattering and determine the oligomeric state.

([11](#))

Data Analysis:

In order to establish the impacts of the G350D mutation, we completed functional studies: GTPase Assay, Mass Photometry, and Negative Stain Electron Microscopy (EM). We began by conducting these functional studies first with the wild-type (WT) protein to see the original function of the protein without the mutation. Afterward, we performed the same studies

using the plasmid DNA with the G350D mutation successfully inserted. We used the WT as a control and compared it to the results of the mutated sequence to determine the effects of the G350D mutation on protein biochemical and structural function.

Results:

Through sequencing, we found that our G350D mutation was successfully inserted into the protein sequence. Originally, the 350th amino acid was glycine (GGA). The mutation changed the glycine to an aspartic acid (GAC), indicating that the mutation was successfully inserted into the plasmid DNA, as shown in Figure 2. We used two primers for sequencing: 740F and Y322A. The 740F primer did not prime, but Y322A successfully worked and provided the results shown in Figure 1.

```
TCCAACCTTATTACCAAATTTGCCACAGAATATTGTAACACTATTGAAGGAAC
template sequence Drp1-1 (Drp1-1.xdna)
TCCNACTTATTACCAAATTTGCCACAGAATATTGTAACACTATTGAAGACACT
aligned sequence Y322AA-Premixed_E08.ab1 (Y322AA-Premixed.seq)
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Figure 2. An image showing the alignment of the Wild Type Drp1 sequence with the sequence primed by the Y322A primer. This alignment shows that the original Drp1 sequence, which contained GGA, was changed to GAC, highlighted in red.

Mass Photometry revealed that the G350D Drp1 forms more hexamer structures than the WT, as highlighted in Figure 3. Contrary to our original expectations of predominantly dimers and possibly monomers, the mutation appears to stabilize larger complexes in solution.

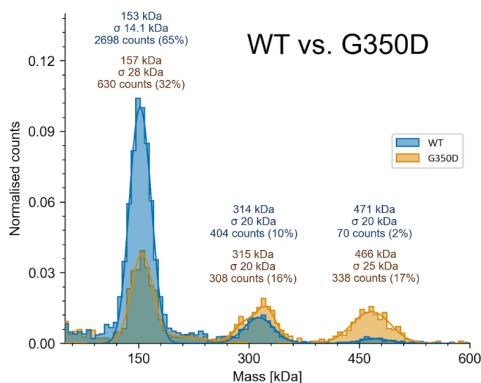


Figure 3. The image depicts the oligomeric forms of both WT and G350D. WT Drp1 predominantly forms dimers, with minimal larger oligomer structures such as hexamers. In contrast, G350D Drp1 shows a more balanced distribution between dimers, tetramers, and hexamers, with significantly more hexamers compared to the WT.

Through the GTPase assay, we found that the G350D mutation exhibits increased rates of enzymatic activity compared to the WT Drp1. The mutated protein had a GTPase activity of 4.15 min^{-1} , while the WT had a lower GTPase activity of 0.53 min^{-1} as highlighted below in Figure 4.

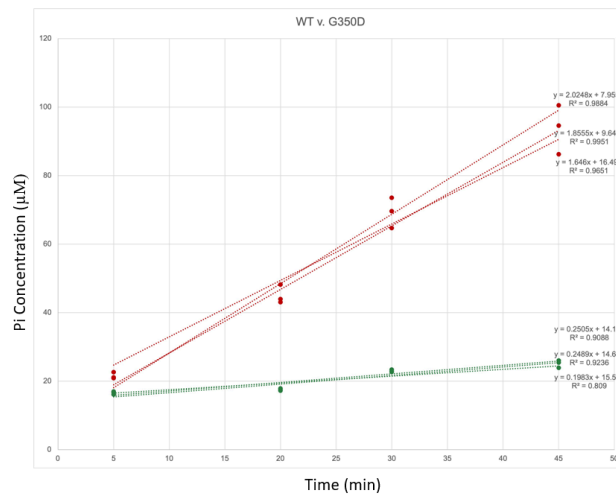


Figure 4. The graph displays the GTPase activity of both WT Drp1 (green) and G350D Drp1 (red). The average slope of the G350D Drp1 samples is steeper, showing an average activity of 4.15 μM phosphate/min/ μM protein, or 4.15 min^{-1} . In contrast, the average slope of the WT Drp1 samples is lower, with an average activity of 0.53 min^{-1} .

Upon conducting negative stain EM, we found that in the presence of GMP-PCP, the WT protein assembled into spiral structures (Figure 5). However, the G350D Drp1 failed to assemble into spirals in the presence of GMP-PCP (Figure 6), showing that it is assembly defective.

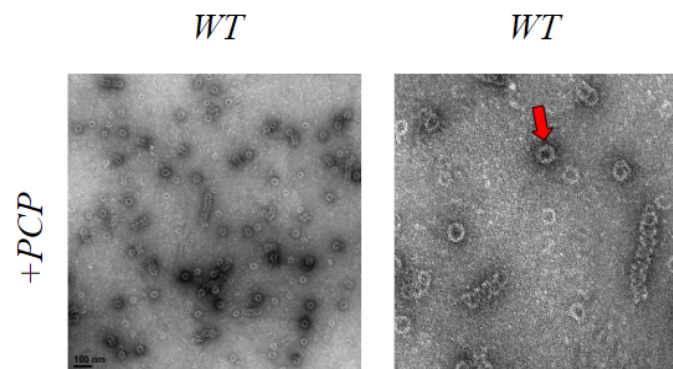


Figure 5. A negative stain EM image of WT Drp1. In the presence of GMP-PCP, Drp1 assembles into spirals as

highlighted by the red arrow. The image on the right is a close-up of the same negative stain EM image of WT shown on the left.

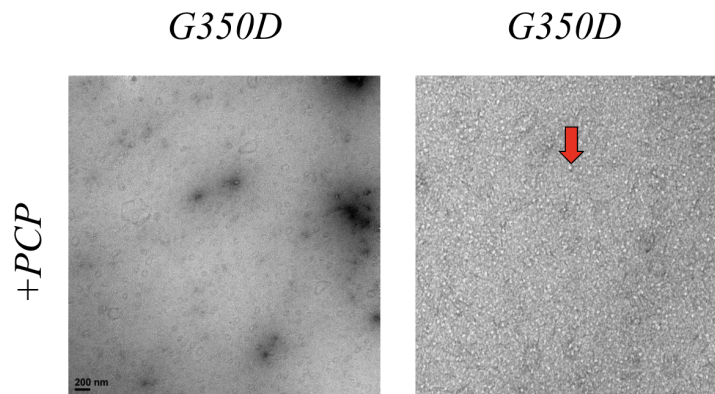


Figure 6. A negative stain EM image of G350D Drp1. In the presence of GMP-PCP, the mutant Drp1 does not assemble into spirals. Instead, it remains unassembled and appears as white dots, as highlighted by the red arrow. The image on the right is a close-up of the same negative stain EM image of WT shown on the left.

Discussion:

Through the results we observed, our hypothesis that the G350D mutations cause an assembly defect was confirmed. Starting with sequencing to validate our sequence change, we used two primers: 740F and Y322A. The 740F primer did not prime, but Y322A successfully worked.

Mass Photometry demonstrated that the G350D mutation assembles into a greater proportion of larger oligomeric states, such as hexamers, compared to the WT. This finding contradicted our original expectation of a prevalence of dimers and possibly even monomers. We initially expected this because we assumed the mutation would result in less assembly, leading to smaller oligomeric states. A previous study showed that proteins forming larger multimers (bigger than dimers) are incapable of mediating fission in cells. ([12](#)) This is likely because Drp1 has to cycle down to a dimer to mediate productive assembly of the fission machinery or the helical polymer that constricts. Thus, our findings from the mass photometry suggest that the

mutation limits assembly differently: the increased proportion of hexamers indicates a shift away from dimers, which would restrict the assembly of the constriction machinery.

GTPase assay results revealed that the G350D mutations exhibit increased enzymatic activity when compared to the WT. The mutated protein had a GTPase activity of 4.15 min^{-1} , while the WT had a lower GTPase activity of 0.53 min^{-1} . One possible explanation for this increased activity is that the G350D mutation causes a conformational change that locks Drp1 into a state that is more catalytically active. The G350D mutation is located in the L1N loop, which is adjacent to the GTPase domain that regulates GTP hydrolysis, a process essential for fission. The increased enzymatic activity we observed could result from a disruption or conformational change in the L1N loop, altering its interactions with the adjacent GTPase domain. This change may have positioned the GTPase domain optimally for GTP hydrolysis, leading to the observed increase in GTPase activity. Furthermore, hexamers may have increased GTPase activity compared to smaller oligomeric forms, possibly also contributing to increased GTPase activity.

Negative stain EM found that in the presence of GMP-PCP, the WT protein assembled into spiral structures. However, the G350D Drp1 failed to assemble into spirals in the presence of GMP-PCP, highlighting an assembly defect and confirming our original hypothesis. This defect is likely due to changes in structural flexibility and charge interactions within the protein. The original amino acid at position 350 is glycine (G), which has a neutral charge and flexible properties that aid loop structures. However, substituting glycine with an aspartic acid (D) introduces a negative charge and reduces the flexibility of the peptide chain at this position, potentially disrupting local interactions within the protein, particularly in the L1N loop. This change could impede Drp1's ability to adapt to the conformational changes necessary for proper

function assembly and could also disrupt interactions with other parts of the protein.

Additionally, the change in charge could influence the assembly behaviors of Drp1.

This study highlights the impacts of the G350D mutation on Drp1 function and structure. Our findings underscore the importance of the L1N loop, which can not only maintain its own structure but can also influence the interactions of adjacent regions. Understanding the impacts of mutations and gaining an understanding of specific sites within Drp1 provides invaluable knowledge of its molecular mechanisms. This knowledge is essential for laying the foundation for developing targeted therapies to mitigate the effects of such mutations, treat neurodevelopmental and neurodegenerative diseases, and restore normal mitochondrial function.

It is also important to note that a limitation of our study was outliers in the GTPase data. While the overall trend for the mutant protein showed a higher rate of increase compared to the wild type, a few time points deviated from this pattern due to pipetting errors. As a result, we excluded those time points from our analysis. Furthermore, this study focused specifically on the G350D mutation, so possible limitations exist. It should be noted that this study has been primarily concerned with isolated proteins. In the future, bioenergetics and cell health should be assessed to complement the studies with isolated proteins to see how interactions are affected. Additionally, the mutation should be incorporated into a cell line to observe how impaired fission would impact Drp1's function. Future research should also study the impacts of other mutations on Drp1 and incorporate further assays to further the collective goal of understanding the mechanisms behind neurodevelopmental and neurodegenerative diseases, ultimately aiding in the development of potential treatments.

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